

Word Scramble!

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How well do you know your math terms? Find out by completing this puzzle! For each line, unscramble the letters to form a well-known math-related word/phrase. Then use the hint below it to locate the letter at that index (0-indexed!). Unscramble these letters to get the final word! Apostrophes are implied where necessary.

1. aonlioswsfce

The second smallest number such that $\phi(n) = 6$, i.e., there are 6 integers less than n that are relatively prime to n .

2. slcaidiraxa

When written in base 3, it is the smallest integer that has six divisors.

3. iosceyyhdikkettinct

A number that is the product of three numbers and the sum of those same three numbers.

4. eaosesmwhtrttrter

The product of 15 consecutive numbers that sum to 90.

5. ncvtroiute

Positive root of $x^3 - 2x^2 - 7x - 4$.

6. lriendyutesiet

A number that is equal to the sum of 3 raised to its digits.

7. incrhalqaytscuzihcuwea

The difference between the first two perfect numbers (positive integers equal to the sum of their proper divisors).

8. rcceimcliucr

$11x$, where x is the expected number of (fair) coin flips needed to get heads.

9. litgreaypahoptner

How many times more subsets can be formed from a set of size 26 than a set of size 23.

10. nltacirgeebso

The number for which the sum and difference of its two prime divisors are both prime.